

DEFINITIONS, TERMINOLOGY, AND INTERPRETATION (NON-LIMITING)

A. Overview

This section defines certain terms used in this disclosure. The definitions are provided for clarity and are not intended to limit the scope of the embodiments. Unless expressly stated otherwise, terms are used in their ordinary meaning as understood by a person skilled in the art.

B. “Node”

As used herein, a “node” refers to a sensing device or sensing subsystem configured to acquire one or more signals and to communicate data to a computing system and/or store data locally. A node may be external, intraluminal, wearable, session-based, continuous-use, or any combination thereof. A node may include one or more sensors, a controller, a power subsystem, memory, and a communications subsystem.

C. “Intimate Physiology”

As used herein, “intimate physiology” refers to physiological signals and patterns associated with intimate-region tissue behavior and/or tissue-adjacent conditions, including signals captured using pressure, strain, magnetic sensing, optical sensing, impedance sensing, dielectric sensing, pH sensing, temperature sensing, motion sensing, contact sensing, and related modalities. “Intimate physiology” includes longitudinal patterns measured over time windows and does not require explicit anatomical imagery.

D. “Privacy-Preserving Outputs”

As used herein, “privacy-preserving outputs” refer to outputs configured to reduce identifiability and stigma by avoiding explicit anatomical imagery and avoiding explicit anatomical labeling. Privacy-preserving outputs may include indices, ranges, zones, curves, trend lines, distributions, confidence indicators, and context tags. Privacy-preserving outputs may be normalized relative to a user baseline or population-referenced models.

E. “Correlation” and “Cross-Device Correlation”

As used herein, “correlation” refers to any computation that relates two or more signals, features, or metrics, including similarity scoring, correlation coefficients, lag analysis, co-variation analysis, sequence detection, distance metrics, and learned similarity metrics. “Cross-device correlation” refers to correlation performed between outputs derived from different nodes.

F. “Cross-User Correlation”

As used herein, “cross-user correlation” refers to correlation performed between privacy-preserving metrics of a first user and privacy-preserving metrics of a second user, under authorization controls. In certain embodiments, cross-user correlation is performed without sharing raw signals between users and without requiring explicit anatomical imagery. Cross-user correlation may support pairing contexts including male-to-female, male-to-male, female-to-female, and other pairings.

G. “Wear Detection,” “Contact Integrity,” and “Artifact Rejection”

As used herein, “wear detection” refers to determining whether a node is being worn or used in a valid context. “Contact integrity” refers to a measure of whether sensor coupling, electrode coupling, optical coupling, and/or pressure/contact coupling meets a validity threshold. “Artifact rejection” refers to identification, down-weighting, or exclusion of segments influenced by motion, misalignment, poor contact, environmental changes, sensor saturation, or other distortions.

H. “Baseline”

As used herein, “baseline” refers to one or more reference measurements, reference distributions, or reference parameters derived from one or more sessions and/or longitudinal windows, used to normalize future metrics, interpret trends, and improve comparability across sessions. Baseline may include geometry references, compliance references, contact references, and/or longitudinal reference distributions.

I. Interpretation Clauses (Non-Limiting)

1. “In various embodiments,” “in certain embodiments,” and similar phrases mean that the described feature may be included in at least one embodiment, and do not require inclusion in all embodiments. 2. “Comprising” is used in an open-ended sense and does not exclude additional elements. 3. “One or more” means one, two, or more. 4. Any numeric range stated herein includes all values within the range, unless expressly stated otherwise. 5. Any list of sensor modalities, form factors, processing modules, or workflows is non-limiting and may be substituted, rearranged, or combined.